

healpix-resample a PSF-Aware HEALPix Resampling for Earth Observation Imagery

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Abstract—Earth observation products such as Sentinel-2 Level-2A images are distributed on local projected grids, whereas large-scale multi-sensor analysis increasingly benefits from global hierarchical indexing systems such as HEALPix. Resampling high-resolution optical imagery onto such grids is, however, not a purely geometric operation: the observed pixels already include an effective point spread function, and conventional interpolation may introduce additional smoothing or scale-dependent artifacts. We present a PSF-aware resampling framework that models Sentinel-2-like UTM observations as sparse linear measurements of a latent HEALPix field. The method estimates the HEALPix representation through a regularized inverse problem and can project the result back to the original UTM grid when compatibility with existing image-processing workflows is required. We validate the approach with two complementary experiments. First, a controlled synthetic detector experiment uses high-resolution ESRI imagery as a latent spatial texture, applies a known Gaussian PSF, and evaluates the ability of the method to recover a consistent HEALPix representation. Second, real Sentinel-2 patches over urban, water, forest, and agricultural scenes are used to quantify UTM-to-HEALPix-to-UTM round-trip fidelity. Image-domain metrics, residual maps, power spectral diagnostics, and runtime measurements show that the proposed formulation preserves spatial information with negligible round-trip error while explicitly accounting for the spatial response of the observation process. The results support the use of PSF-aware HEALPix resampling as a practical component for resolution-preserving Earth observation data harmonization.

Index Terms—Discrete Global Grid Systems, DGGS, HEALPix, Sentinel-2, point spread function, resampling, regridding, inverse problems, Earth observation, UTM.

I. INTRODUCTION

Earth observation (EO) data processing depends on a broad range of local map projections, product tiling schemes, and sensor-specific sampling geometries. Sentinel-2 Level-2A products, for example, are delivered as ortho-images in the UTM/WGS84 cartographic system, with spectral bands available at ground sampling distances of 10, 20, or 60 m depending on the band [1], [2]. Although such projected products are convenient for operational distribution and regional processing, spatial discretization is not geometrically neutral. Differences in cell area, shape, orientation, and neighbourhood structure can affect spatial analyses, including at local scales.

A unified indexing framework is therefore useful for global, multi-resolution, or multi-sensor workflows, but its geometric properties and distortions must be made explicit [3].

Discrete Global Grid Systems (DGGS) address this challenge by dividing the Earth's surface into hierarchical cells with explicit indices and multiple resolutions [3], [4]. HEALPix, originally developed for the analysis of spherical astronomical data, is especially attractive in this context because it provides a hierarchical structure, equal-area cells, and iso-latitude sampling [5]. These characteristics are advantageous for EO data cubes, where global indexing, scale changes, and area-consistent statistical analyses are required [3]. However, a DGGS also imposes a discrete set of admissible cell sizes. In practice, the chosen HEALPix resolution does not necessarily match the native sampling scale of a given sensor. For instance, a Sentinel-2 10 m band may be represented on a finer HEALPix grid, such as a grid with cells of approximately 6.5 m in the present experiments. This mismatch is not merely a storage or indexing detail: estimating values on a grid finer than the native detector sampling raises the question of what spatial information can actually be reconstructed from the observation.

Transferring high-resolution imagery from a projected raster grid to a spherical DGGS is therefore not a neutral operation. Conventional raster resampling methods, such as nearest-neighbor interpolation, bilinear interpolation, cubic convolution, or locally weighted interpolation, are typically formulated as geometric interpolation schemes, in which the value assigned to a target-grid cell is inferred from source-grid samples selected according to their projected spatial coordinates. In isolation, these methods do not incorporate an explicit representation of the sensor point spread function (PSF), nor do they solve the associated observation inverse problem [6]–[8].

In practice, each pixel value is generated by an image-formation process characterized by a finite spatial response. This response encompasses optical blurring, detector integration, platform motion, atmospheric propagation, and any resampling operations applied during product generation. It is typically represented either in the spatial domain by an effective point spread function (PSF) or in the frequency domain by a modulation transfer function (MTF). For Sentinel-2 MSI, the MTF is explicitly monitored as part of the product quality-control framework, with in-orbit estimates derived from sharp edges or line features [9]. Neglecting this effective spatial response during resampling alters the spatial transfer characteristics of the product: it can introduce uncontrolled

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smoothing, distort spatial power spectra, and obscure the distinction between a grid that represents the original observation and one that represents an additionally degraded product [10], [11]. This limitation is not specific to HEALPix or Sentinel-2. Inamdar et al. [10], in the context of hyperspectral image resampling, demonstrated that disregarding the sensor spatial point spread function can lead to unrealistic spatial degradation, and proposed a resampling strategy that explicitly incorporates the spatial response.

When the target grid is finer than the native sampling scale, PSF-aware resampling becomes closely related to classical image restoration and super-resolution. The aim is then to estimate a latent image or field on a finer grid from observations that are blurred, sampled, and noisy. This class of problems has a long history, from Richardson–Lucy deconvolution with a known point spread function [12], [13] to model-based super-resolution methods that explicitly combine geometric warping, blur, downsampling, and noise in a forward observation model [14]–[16]. Such inverse problems are intrinsically ill posed: a finer target grid contains degrees of freedom that are only partially constrained by the data. Components that are strongly coupled across scales by the image-formation model may be partially recovered, whereas weakly coupled components, such as nearly white noise or unresolved high-frequency variability, can be amplified if the inverse problem is left unconstrained. Regularization is therefore not an implementation detail but a necessary part of the formulation, as in classical super-resolution and deconvolution methods [14]–[16].

This paper presents a PSF-aware methodology for resampling gridded remote-sensing imagery onto HEALPix tessellations. The proposed framework is sensor-agnostic: it assumes only that the input data are provided on a known projected grid and that an effective spatial-response model is available. Sentinel-2 data are used as a representative real-world test case, without restricting the approach to this particular sensor. The same formalism is applicable to any gridded observation for which both geolocation information and an effective spatial-response function can be specified. In this sense, the method should be understood as a controlled inverse resampling framework rather than as a generic interpolation operator.

We address two complementary validation objectives. The first pertains to physical consistency under a controlled spatial-response model. To this end, we construct a synthetic detector experiment in which a patch from the Esri World Imagery basemap is employed as a high-resolution latent spatial texture, subsequently convolved with a known Gaussian PSF, and then reconstructed using the proposed inverse formulation [17]. This experiment makes it possible to separate the latent texture, the PSF-blurred observation, and the reconstructed HEALPix field. The second objective concerns practical reversibility on real projected remote-sensing data. For this purpose, we use Sentinel-2 imagery distributed on UTM/MGRS tiles as a representative instance of high-resolution optical EO data defined on a locally projected grid [18]–[20]. Because an unblurred radiance ground truth is not available for real observations, this experiment does not attempt to demonstrate exact recovery of an unknown

unblurred scene. Instead, it assesses whether the estimated HEALPix representation can be reprojected back onto the original UTM/MGRS grid with negligible numerical loss under the tested configurations.

The use of a finer DGGS grid may increase the number of stored samples relative to the native raster. This paper does not attempt to optimize storage layout, compression, or multi-resolution encoding. However, the storage question must be considered together with the redundancy of existing projected tiling systems. In the case of Sentinel-2, the UTM/MGRS tiling grid contains systematic overlaps and duplicated coverage; Bauer-Marschallinger and Falkner [21] reported that this grid-inherent redundancy inflates global land remote-sensing datasets by about one third. A global DGGS representation may therefore trade a controlled increase in sampling density for a simpler, non-overlapping global indexing structure. Quantifying this trade-off for large-scale archives is beyond the scope of the present work and is left for future study.

The main contributions of this work are:

- a PSF-aware projected-grid-to-HEALPix resampling scheme based on a sparse linear observation model of an underlying latent HEALPix field;
- a formulation that explicitly handles the case where the target HEALPix grid is finer than the native sensor sampling, making the connection with regularized super-resolution and deconvolution;
- a controlled synthetic-detector framework that separates the latent spatial texture, the PSF-blurred observation, and the reconstructed field;
- a Sentinel-2 round-trip validation, used as a representative real-data test case, showing negligible numerical loss when mapping the inferred HEALPix representation back to the original UTM/MGRS grid under the tested conditions;
- a multiscale spectral diagnostic to quantify how spatial variability is preserved, attenuated, or amplified across scales;
- an open-source `healpix-resample` package designed for HEALPix/DGGS-based Earth-observation data cube processing pipelines.

II. BACKGROUND AND RELATED WORK

A. HEALPix as a Latent Spherical Support

The HEALPix tessellation divides the sphere into $12N_{\text{side}}^2$ equal-area cells, with $N_{\text{side}} = 2^L$ at hierarchy level L [5]. Its equal-area, hierarchical, and iso-latitude properties make it useful for area-consistent statistics, multi-resolution indexing, and efficient spherical data access.

In this work, HEALPix is used as a latent spherical support rather than as a sensor-specific sampling grid. The proposed method does not depend on a particular sensor resolution or on a fixed HEALPix level. In the experiments reported here, Sentinel-2 imagery is used only as a representative high-resolution optical test case, and we choose $L = 20$, corresponding to $N_{\text{side}} = 1,048,576$ and an equivalent cell size of approximately 6.2–6.5 m at the Earth’s surface. This choice makes the target support finer than the native 10 m

sampling of the Sentinel-2 bands considered in the test case, but the same formulation applies to coarser or finer sensors and to other choices of DGGS resolution.

When the target DGGS grid is finer than the native sampling scale, the resampling problem has more degrees of freedom than are directly observed. The finer grid should therefore not be interpreted as containing independent information at the target-cell scale by construction. Instead, these additional degrees of freedom must be constrained by the forward observation model, the effective spatial response, and regularization, as in image restoration and super-resolution problems.

B. Resampling as an Observation Model

In this work, resampling is formulated as an observation model rather than as a purely geometric interpolation. The projected image is treated as a set of spatially weighted measurements of a latent HEALPix field, with weights determined by geolocation and by an assumed effective PSF. This connects the method to classical inverse formulations used in deconvolution, super-resolution, and model-based image reconstruction [12], [13], [15], [22], [23].

The objective is not blind super-resolution from a single image, but controlled inverse resampling: the recovered HEALPix field is explicitly tied to the assumed observation operator and constrained by regularization.

III. METHOD

A. Forward Observation Model

Let $\mathbf{y} \in \mathbb{R}^N$ denote the vectorized input image patch on a projected source grid, where N is the number of valid source pixels. Let $\mathbf{x} \in \mathbb{R}^M$ denote the latent field defined on the M HEALPix cells intersecting the patch and the finite support of the assumed spatial response. In the Sentinel-2 experiment, the source grid is the native UTM/MGRS product grid, but the formulation itself is not tied to this projection.

We model the source-grid observations as

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}, \quad (1)$$

where $\mathbf{A} \in \mathbb{R}^{N \times M}$ is a sparse observation operator and $\mathbf{n}$ gathers measurement noise, modeling errors, geolocation uncertainties, and mismatch between the assumed and true spatial response. Conceptually, each row of $\mathbf{A}$ discretizes the image-formation integral that maps the latent spherical field to one observed source-grid pixel.

For each source pixel i , nearby HEALPix cell centers $j \in \mathcal{N}(i)$ are selected within a finite support radius around the pixel footprint. The non-zero entries of $\mathbf{A}$ are defined as normalized spatial-response weights,

$$A_{ij} = \frac{K(d_{ij}; \sigma_{PSF})}{\sum_{k \in \mathcal{N}(i)} K(d_{ik}; \sigma_{PSF})}, \quad (2)$$

where d_{ij} is the local projected or geodesic distance between source pixel i and HEALPix cell j , and σ_{PSF} controls the width of the assumed effective PSF. In the experiments, K is chosen as a Gaussian kernel, with FWHM = $2\sqrt{2 \ln 2} \sigma_{PSF}$. The row normalization ensures that a spatially constant latent

Algorithm 1 PSF-aware projected-grid-HEALPix resampling.

- 1: Input: projected source image $\mathbf{y}$, geolocation information, HEALPix level L , PSF width σ_{PSF} , regularization parameter λ .
 - 2: Select all HEALPix cells intersecting the image patch and the finite PSF support.
 - 3: For each valid source pixel, query neighboring HEALPix cells.
 - 4: Compute normalized spatial-response weights and assemble the sparse observation operator $\mathbf{A}$.
 - 5: Solve Eq. (3) for the latent field $\hat{\mathbf{x}}$.
 - 6: Optionally compute the source-grid prediction $\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}$.
 - 7: Output: HEALPix cell identifiers, estimated HEALPix values, optional source-grid prediction, and residual diagnostics.
-

field is mapped to a spatially constant observation, while the finite support keeps the operator sparse.

This construction should be understood as a practical discretization of an effective spatial response, not as a full radiometric model of the instrument. More detailed versions of $\mathbf{A}$ could include anisotropic or band-dependent PSFs, detector footprints, scan-direction effects, terrain corrections, or uncertainty weights. The present work focuses on the consequences of introducing an explicit spatial-response operator into the resampling step.

B. Regularized Inversion

The latent HEALPix field is estimated by solving the regularized inverse problem

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda \mathcal{R}(\mathbf{x}), \quad (3)$$

where $\mathcal{R}$ is a stabilizing prior and λ controls the balance between observation fidelity and regularity of the latent field. In the current implementation, $\mathcal{R}$ is a Tikhonov-type penalty, but the same framework can accommodate local smoothness, sparsity, anisotropic, or multi-resolution priors.

Regularization is essential when the HEALPix support is finer than the native sampling scale of the source image. In that case, the latent field contains degrees of freedom that are only weakly constrained by the observations. Without stabilization, the inversion may amplify noise or unresolved high-frequency variability. The estimated field $\hat{\mathbf{x}}$ should therefore be interpreted as the regularized latent field most consistent with the assumed observation operator, not as an unconstrained recovery of independent sub-pixel information.

For diagnostic purposes, the estimated field can be mapped back to the source observation space as

$$\hat{\mathbf{y}} = \mathbf{A}\hat{\mathbf{x}}. \quad (4)$$

This quantity is the source-grid prediction implied by the recovered HEALPix field and the same spatial-response model. It is used to test round-trip consistency: if $\hat{\mathbf{y}}$ closely matches $\mathbf{y}$, then the HEALPix representation remains compatible with the original observation space under the assumed model.

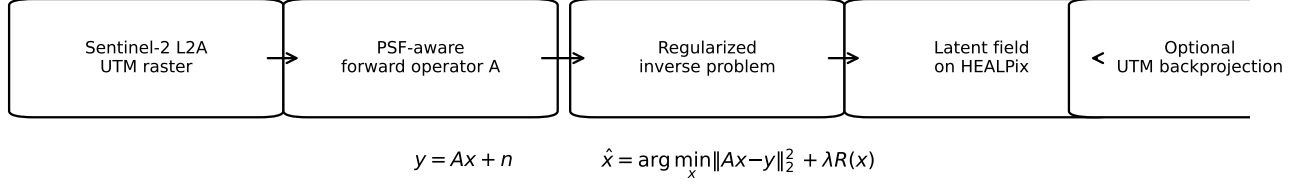


Fig. 1. PSF-aware projected-grid-HEALPix resampling workflow. The projected source image is modeled as a sparse set of spatially weighted measurements of a latent HEALPix field. The recovered field can be stored directly in a DGGS representation or mapped back to the source grid to evaluate consistency with the original observation space.

IV. VALIDATION STRATEGY

A. Overview

The validation is designed to separate two questions that are often conflated in resampling studies. The first question is whether the proposed operator is physically consistent with a prescribed spatial-response model. This requires a controlled experiment in which a latent field and a degraded observation are both available. The second question is whether the resulting HEALPix representation remains compatible with real projected observations. This requires a round-trip experiment on real data, but it cannot by itself demonstrate recovery of an unknown unblurred scene.

We therefore use two complementary validation settings. The controlled experiment uses synthetic detector observations generated from high-resolution image textures. The real-data experiment uses Sentinel-2 as a representative example of a high-resolution optical EO product distributed on a projected UTM/MGRSgrid. Sentinel-2 is used here as a test case, not as a restriction of the method.

B. Controlled Synthetic Detector Experiment

The controlled experiment uses four benchmark scene types: urban, water, forest, and agriculture. For each location, an image patch from the Esri World Imagery basemap is used as a high-resolution spatial texture. This image is not interpreted as a Sentinel-2 radiometric truth; it is used only as a controlled latent texture from which a simulated detector observation can be generated [17]. A Gaussian PSF with prescribed full width at half maximum is applied to this texture, producing a PSF-degraded observation. The proposed PSF-aware resampler is then applied to this simulated observation, and the recovered HEALPix field is compared both with the original latent texture and with the PSF-degraded reference.

This design isolates three quantities that are entangled in real satellite observations: the latent spatial texture, the spatial response of the measurement process, and the reconstructed grid representation. It is therefore the experiment used to assess the behaviour of the inverse formulation and the role of regularization. Since the target HEALPix grid may be finer than the native simulated detector scale, this test also probes the super-resolution regime, where some degrees of freedom of

the latent field are only weakly constrained by the observation operator.

C. Real Projected-Grid Round-Trip Experiment

The second experiment evaluates practical reversibility on real projected remote-sensing data. We use Sentinel-2 image patches over the same four scene types. Because the unblurred radiance field is not available for real observations, this experiment is not an absolute deconvolution benchmark. Instead, it evaluates whether the estimated HEALPix representation preserves the information required to predict the original source-grid observation under the same observation model.

Let y denote the input projected-grid image, $\hat{x}$ the estimated HEALPix field, and $\hat{y} = A\hat{x}$ the corresponding source-grid prediction. The round-trip residual is defined as

$$r = \hat{y} - y. \quad (5)$$

Small residuals indicate that the HEALPix representation remains compatible with the original observation space. They do not imply that the real instrumental PSF has been inverted.

D. Metrics

Image-domain fidelity is assessed using root mean square error (RMSE), mean absolute error (MAE), bias, Pearson correlation, coefficient of determination R^2 , peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM) [24]. These metrics summarize global agreement, but they do not indicate which spatial scales are preserved, attenuated, or amplified.

We therefore also compute isotropically averaged one-dimensional power spectra. For a reference spectrum $P_{\text{ref}}(k)$ and an estimated spectrum $P_{\text{est}}(k)$, we define the normalized spectral discrepancy

$$E_{\text{PSD}}(\mathcal{K}) = \frac{\sum_{k \in \mathcal{K}} |P_{\text{est}}(k) - P_{\text{ref}}(k)|}{\sum_{k \in \mathcal{K}} P_{\text{ref}}(k) + \epsilon}, \quad (6)$$

where $\mathcal{K}$ denotes a selected range of spatial frequencies and ϵ prevents numerical instability for low-power spectra. In the synthetic experiment, P_{ref} can refer either to the latent no-PSF texture or to the PSF-degraded observation, depending on whether the goal is to evaluate latent-field recovery or observation consistency. In the real-data round-trip experiment, the relevant reference is the input projected-grid observation.

V. RESULTS

A. Controlled Synthetic Detector Recovery

Fig. 2 shows the controlled synthetic detector experiment for the four benchmark scenes. The real Sentinel-2 patches are shown only as contextual examples of the scene types. The Esri no-PSF images provide the latent spatial textures used to generate the simulated observations. The corresponding PSF-degraded images represent the synthetic detector measurements, and the final row shows the HEALPix estimates rendered in the local view.

Table I reports representative synthetic-detector metrics over common HEALPix cells. The key result is that, for the current configuration, the recovered field is generally much closer to the PSF-degraded reference than to the no-PSF latent texture. For example, in the urban case, the RMSE with respect to the no-PSF texture is 0.1260, whereas the RMSE with respect to the PSF-degraded reference is 0.0105. Similar behaviour is observed for the water and forest scenes. This indicates that the present configuration primarily recovers a field that is consistent with the assumed observation process, rather than performing an aggressive deconvolution toward the original latent texture.

This result is important for the interpretation of the method. Since the HEALPix support may be finer than the native observation scale, the inverse problem contains degrees of freedom that are not independently constrained by the data. The regularized solution therefore favours spatial components that are supported by the observation operator and by the prior, while suppressing or attenuating weakly constrained high-frequency components. This is the expected behaviour of a stable inverse resampling scheme. It should not be interpreted as a failure to produce blind super-resolution, but rather as evidence that the current regularization prevents uncontrolled amplification of unresolved structures.

The agriculture scene shows a slightly different behaviour: the estimate is marginally closer to the no-PSF texture than the PSF-degraded observation is, with RMSE decreasing from 0.0167 to 0.0144. This suggests that partial recovery of spatial contrast can occur when the scene texture, the assumed PSF, and the regularization are favourable. However, this effect is scene-dependent and should be quantified with frequency-resolved diagnostics before being presented as general PSF correction.

B. Spectral Behaviour

The image-domain metrics in Table I provide a compact summary of global agreement, but they do not reveal how the inversion behaves across spatial scales. Fig. 3 therefore compares one-dimensional power spectra for the latent no-PSF texture, the PSF-degraded observation, and the recovered HEALPix field.

The spectra provide a scale-dependent interpretation of the recovery. At low spatial frequencies, the no-PSF texture, the PSF-degraded observation, and the estimate are expected to be close, because these scales are transmitted efficiently by the spatial-response model. At high spatial frequencies, the PSF attenuates the observation, and the inversion becomes

increasingly sensitive to regularization. In this regime, a stable method should avoid amplifying components that are weakly constrained by the data, especially components whose statistics are close to white noise. The observed high-frequency behaviour therefore provides a direct diagnostic of the trade-off between spatial sharpening and noise amplification.

Fig. 4 compares the spectra of real Sentinel-2 patches and synthetic Esri-derived detector images. This diagnostic is not intended to establish radiometric equivalence between Esri and Sentinel-2. Instead, it checks whether the synthetic textures provide spatial-frequency content representative of the scene classes used in the real-data experiment. Such a comparison is useful because the relevance of the controlled experiment depends on the presence of spatial structures at scales comparable to those encountered in the real observations.

Overall, the spectral diagnostics support a conservative interpretation. The current implementation preserves the large- and intermediate-scale structure transmitted by the assumed PSF and regularization. Evidence for true recovery of attenuated high-frequency content is scene-dependent and should be assessed using explicit frequency-band error metrics such as Eq. (6).

C. Real Sentinel-2 Round-Trip Consistency

Fig. 5 shows the real-data round-trip experiment. The original projected-grid image and the source-grid prediction obtained from the HEALPix representation are visually almost indistinguishable. The signed and absolute residual maps show that the remaining differences are concentrated near high-gradient features, such as roads, field boundaries, shorelines, and forest edges. This spatial distribution is expected for a finite-support observation operator and is more informative than global error metrics alone.

Table II reports representative round-trip metrics. The RMSE values are very small, correlations are close to unity, and SSIM values are also close to unity. These results show that, under the tested configurations, the HEALPix representation retains the information required to reproduce the original Sentinel-2 projected-grid observation. This is a consistency result: it demonstrates that storing and manipulating the data through the HEALPix representation does not introduce a practically significant numerical loss in the tested cases.

The round-trip result should be interpreted together with the synthetic experiment. The real-data experiment demonstrates that the HEALPix representation is compatible with the input observation space. The synthetic experiment provides the only setting in which one can separately evaluate consistency with a known degraded observation and proximity to a latent no-PSF texture. Together, they show that the method is stable, observation-consistent, and suitable for representing projected EO imagery on a HEALPix support, while leaving stronger claims of PSF correction to controlled frequency-band analyses.

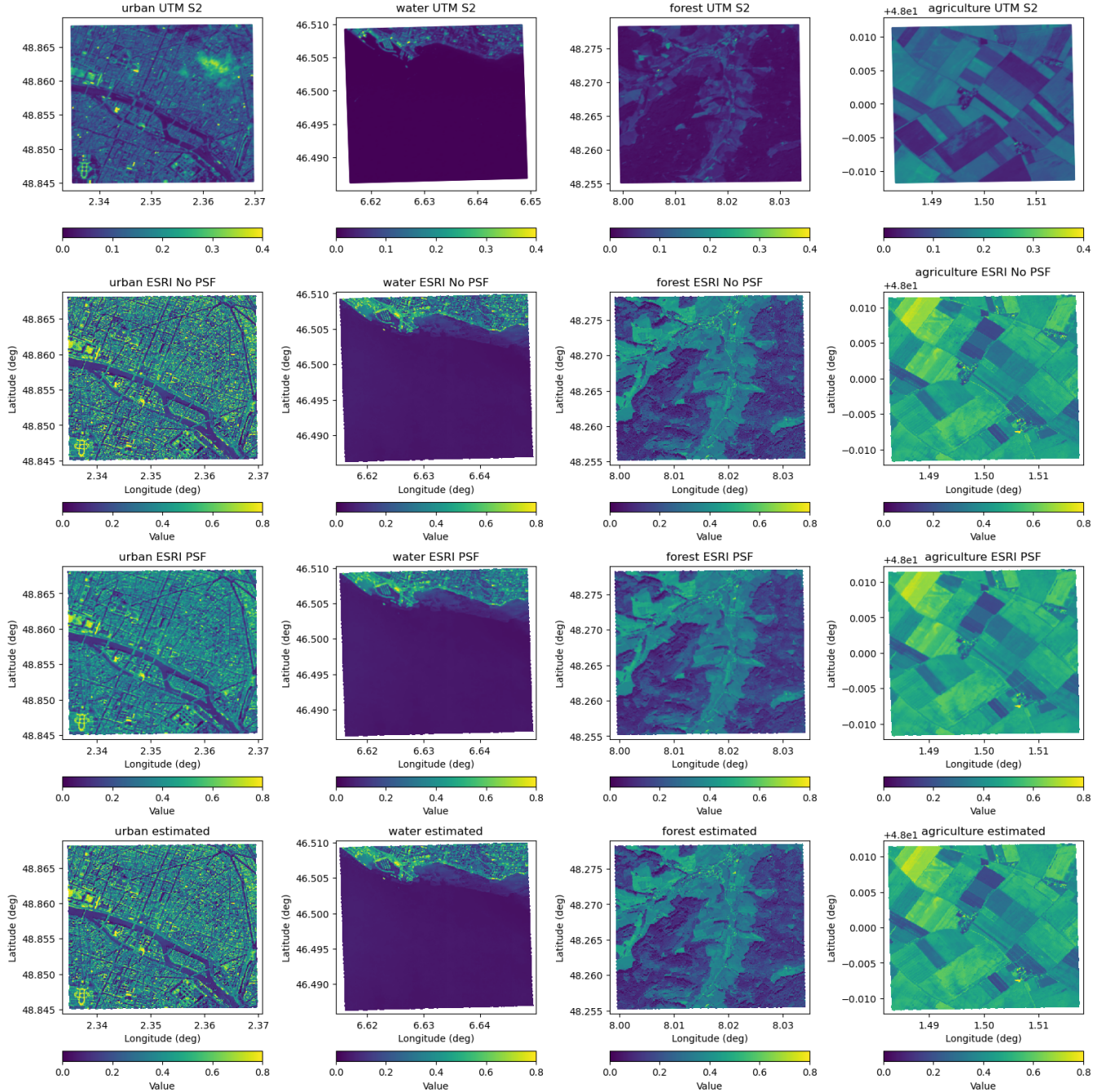


Fig. 2. Controlled synthetic detector experiment for four benchmark scenes. The Esri no-PSF image is used as a latent spatial texture, the Esri PSF image is the simulated detector observation, and the estimated image is the PSF-aware HEALPix recovery rendered in the local view. Sentinel-2 patches are included only to illustrate the corresponding scene types.

VI. DISCUSSION

A. Observation Consistency Versus Deconvolution

The experiments support a precise interpretation of the proposed method. The method is PSF-aware because the spatial response is included explicitly in the sparse observation operator. It is also observation-consistent because the recovered HEALPix field can be mapped back to the source grid with very small residuals in the real Sentinel-2 experiment. However, these two properties are not equivalent to blind deconvolution.

On real Sentinel-2 data, the unblurred radiance field is unknown. A round-trip experiment can therefore show compatibility with the original observation, but it cannot prove that the true instrumental PSF has been inverted. The controlled synthetic experiment provides a more rigorous test of the inverse formulation because the latent texture and the PSF-degraded observation are both available. In the current configuration, the recovered field is generally closer to the PSF-degraded observation than to the no-PSF texture. This indicates that the method behaves primarily as a stable, PSF-

TABLE I
SYNTHETIC DETECTOR METRICS OVER COMMON HEALPIX CELLS. THE VALUES SUMMARIZE THE CURRENT CONTROLLED EXPERIMENT AND ARE INTERPRETED AS DIAGNOSTICS OF OBSERVATION CONSISTENCY AND PARTIAL LATENT-FIELD RECOVERY.

Scene	Comparison	RMSE	MAE	Corr.	R^2
Urban	Estimated vs. Esri no-PSF	0.1260	—	—	—
Urban	Estimated vs. Esri PSF	0.0105	—	—	—
Water	Estimated vs. Esri no-PSF	0.0396	—	—	—
Water	Estimated vs. Esri PSF	0.0039	—	—	—
Forest	Estimated vs. Esri no-PSF	0.0430	—	—	—
Forest	Estimated vs. Esri PSF	0.0047	—	—	—
Agriculture	Esri PSF vs. Esri no-PSF	0.0167	0.0093	0.9880	0.9757
Agriculture	Estimated vs. Esri no-PSF	0.0144	0.0083	0.9910	0.9820
Agriculture	Estimated vs. Esri PSF	0.0141	0.0083	0.9914	0.9815

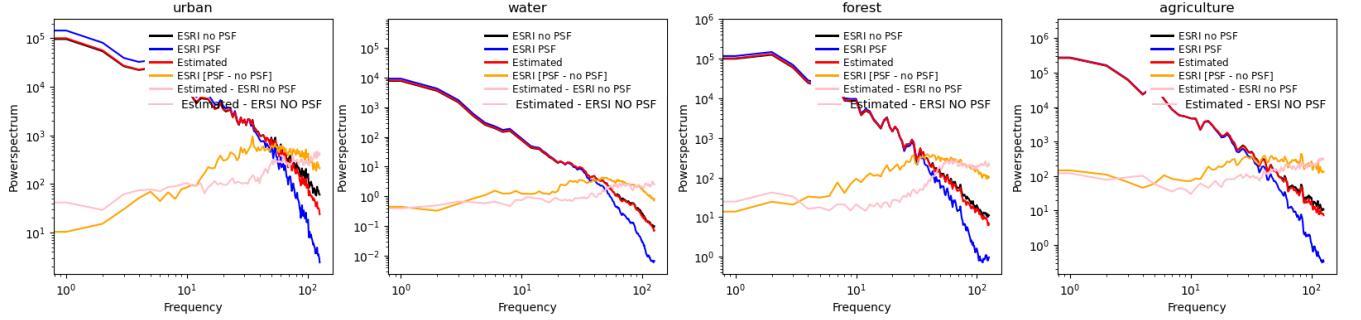


Fig. 3. Power spectral diagnostics for the synthetic detector experiment. Spectra are shown for the Esri-no-PSF texture, the PSF-degraded reference, and the estimated HEALPix field. The comparison reveals the spatial-frequency ranges where the recovered field follows the latent texture, the degraded observation, or departs from both.

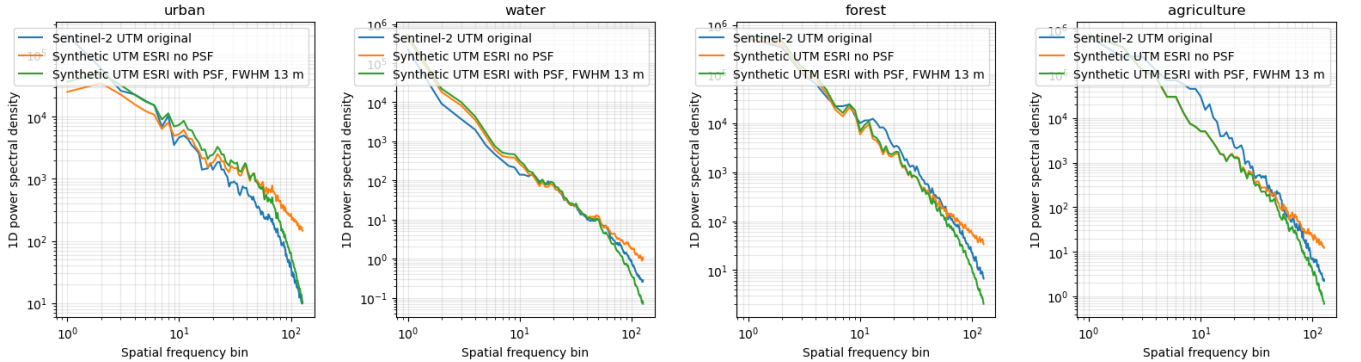


Fig. 4. Comparison of power spectra for real Sentinel-2 projected-grid patches and synthetic Esri-derived detector images. The diagnostic evaluates whether the controlled textures contain spatial frequencies representative of the scene types used in the round-trip experiment, without interpreting Esri imagery as Sentinel-2 radiometric truth.

aware inverse resampling scheme rather than as a strong deconvolution method.

This behaviour is scientifically useful. In DGGS applications, the first requirement is often to produce a global, indexed representation that remains traceable to the original observation. A method that explicitly preserves the observation model is preferable to a purely geometric interpolation whose effective transfer function is uncontrolled. Stronger PSF correction may be possible, but it requires careful tuning of the spatial-response model, the target HEALPix level, and the regularization.

B. Relation to Super-Resolution

The experiments also illustrate the connection between PSF-aware DGGS resampling and super-resolution. When the target HEALPix cell size is smaller than the native sampling scale, the latent field has more degrees of freedom than the original observation. Some of these degrees of freedom are constrained by the spatial-response model and by spatial correlations in the image. Others, especially high-frequency components with weak coupling to the observation, are not reliably recoverable from a single image.

Regularization is therefore central to the method. A weakly regularized inversion may increase apparent sharpness but also amplify noise or unresolved variability. A strongly regularized inversion may remain stable but stay close to the PSF-

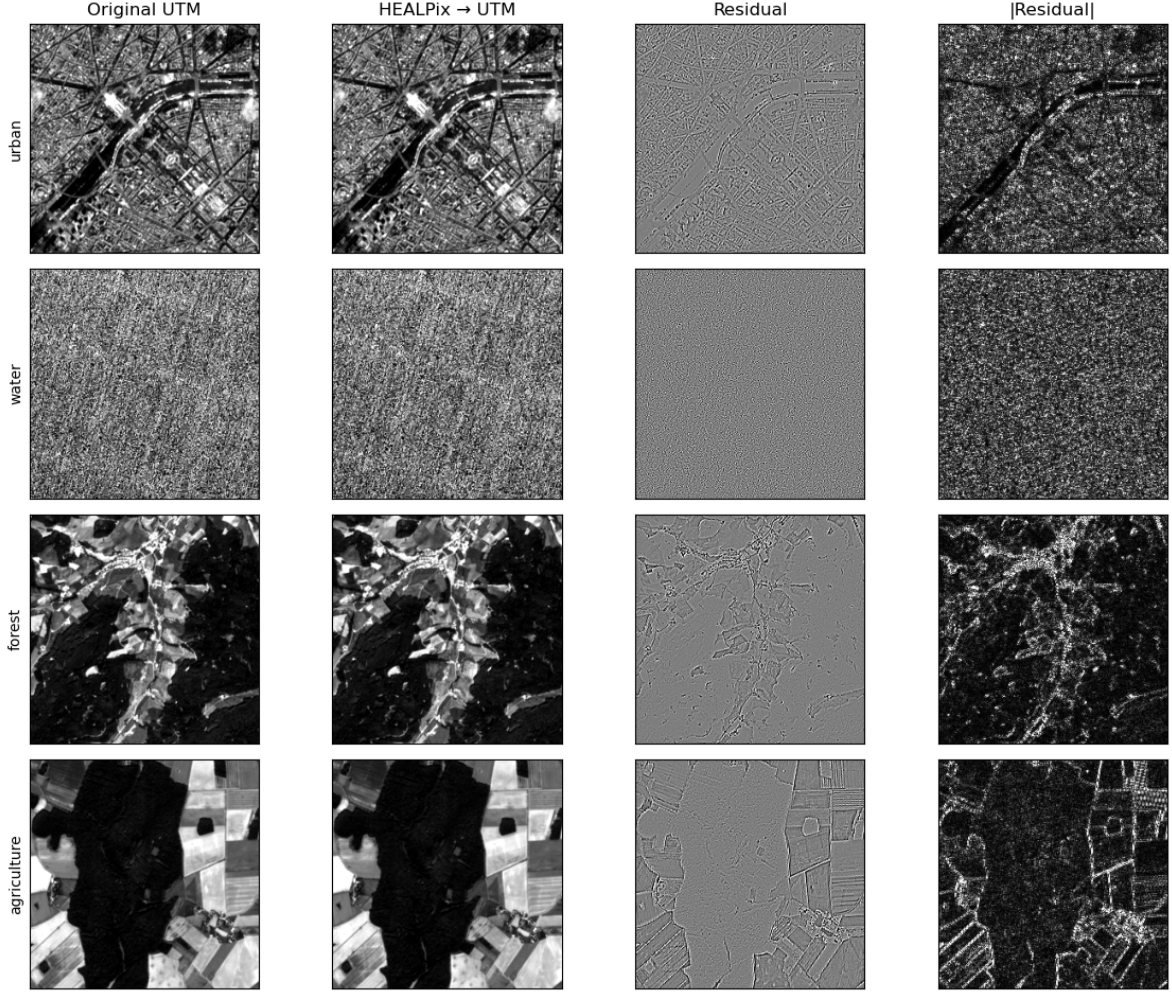


Fig. 5. Real Sentinel-2 projected-grid-HEALPix-projected-grid round-trip experiment. Columns show the original projected-grid patch, the source-grid prediction reconstructed from the HEALPix estimate, the signed residual, and the absolute residual. This experiment evaluates practical reversibility, not absolute deconvolution, because an unblurred Sentinel-2 truth is unavailable.

TABLE II

REPRESENTATIVE SENTINEL-2 PROJECTED-GRID-HEALPIX-PROJECTED-GRID ROUND-TRIP METRICS FOR THE FOUR BENCHMARK SCENES. THE METRICS QUANTIFY CONSISTENCY WITH THE ORIGINAL PROJECTED-GRID OBSERVATION, NOT RECOVERY OF AN UNKNOWN UNBLURRED RADIANCE FIELD.

Scene	PSF-aware RMSE	Reference smoothing RMSE	PSF-aware Corr.	PSF-aware SSIM
Urban	8.8×10^{-5}	2.24×10^{-2}	≈ 1	0.999998
Water	2.0×10^{-6}	4.17×10^{-4}	≈ 1	0.999996
Forest	2.4×10^{-5}	6.35×10^{-3}	≈ 1	0.999996
Agriculture	2.2×10^{-5}	5.83×10^{-3}	≈ 1	0.999999

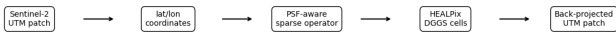


Fig. 6. Round-trip RMSE for the four Sentinel-2 benchmark scenes. The small values indicate that the HEALPix representation remains highly consistent with the original projected-grid observation under the tested settings.

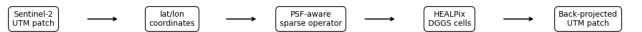


Fig. 7. Round-trip SSIM for the four Sentinel-2 benchmark scenes. Values close to unity indicate practical preservation of image structure after estimation on the HEALPix grid and prediction back to the original projected grid.

degraded observation. The current results are consistent with

the latter regime. They demonstrate stability and round-trip

consistency, while showing only limited and scene-dependent evidence of recovery toward the no-PSF latent texture. Future work should quantify this trade-off through systematic sweeps over PSF width, target HEALPix level, and regularization strength.

C. Spectral Interpretation

Power spectral diagnostics are essential because global image metrics can hide scale-dependent effects. RMSE, correlation, and SSIM may indicate excellent agreement even when high-frequency content has been attenuated or amplified. By contrast, one-dimensional spectra make it possible to identify the frequency ranges in which the recovered field follows the latent texture, the PSF-degraded observation, or neither.

The current spectral results suggest that low and intermediate spatial frequencies are well preserved. The high-frequency regime is more sensitive to scene texture and regularization, which is expected for an inverse problem on a finer target grid. A stronger claim of partial PSF correction should therefore be made only if the recovered field is shown to reduce the spectral discrepancy with respect to the no-PSF latent texture over a defined frequency band, while avoiding amplification in bands dominated by noise or unresolved variability.

D. Baselines and Limitations

The present results focus on the proposed PSF-aware formulation and on internal consistency diagnostics. A full benchmark against conventional resampling methods remains necessary for a final performance comparison. Relevant baselines include nearest-neighbor interpolation, bilinear interpolation, cubic convolution, inverse-distance weighting, and conservative or area-weighted resampling when applicable. These baselines should be evaluated with the same image-domain and spectral metrics, and with the same target HEALPix resolution.

The current validation also uses four benchmark scenes. This is sufficient to demonstrate the behaviour of the method across contrasting spatial textures, but it does not exhaust the diversity of EO scenes. Additional cases such as ocean, snow and ice, clouds, mountains, and heterogeneous coastal regions would strengthen the assessment. These scenes should be included only when the texture sampling, masking, and metric computation are robust for all cases.

Several implementation choices also affect the reported residuals and should be documented explicitly in the final manuscript: the finite support radius of the PSF kernel, the treatment of empty or weakly constrained HEALPix cells, the handling of masks and invalid pixels, boundary effects at the edge of the patch, and the numerical solver used for the regularized inverse problem. These choices are not secondary details; they determine the conditioning of the problem and the interpretation of the recovered high-frequency content.

E. Storage and Data-Cube Implications

Using a HEALPix grid finer than the native sensor sampling may increase the number of stored samples. This paper does not optimize compression, multi-resolution encoding, or storage layout. Nevertheless, the storage cost should be considered

in the broader context of EO data-cube design. A global DGGS representation can reduce dependence on local tiling systems and avoid some forms of overlap and duplication present in projected tiling schemes. The relevant question is therefore not only whether a finer HEALPix grid contains more cells than a local raster patch, but whether the resulting global representation provides a better trade-off between indexing simplicity, geometric consistency, redundancy, compression, and analysis-readiness.

VII. CONCLUSION

We introduced a PSF-aware framework for resampling projected remote-sensing imagery onto HEALPix grids. The method formulates resampling as a sparse inverse problem in which the source image is modeled as spatially weighted measurements of a latent HEALPix field. This formulation makes the spatial response explicit and provides a controlled alternative to purely geometric interpolation.

The validation separates two complementary objectives. A controlled synthetic detector experiment evaluates the behaviour of the inverse formulation when a latent texture and a known PSF-degraded observation are both available. A real Sentinel-2 round-trip experiment evaluates whether the estimated HEALPix representation remains compatible with an operational projected-grid observation. The results show negligible numerical loss in the tested Sentinel-2 round trips and stable recovery in the synthetic detector setting.

The current evidence supports describing the method as PSF-aware, regularized, and observation-consistent. It does not yet justify a general claim of blind super-resolution or full PSF deconvolution from a single real observation. Instead, the results show that spatial-response-aware resampling can preserve traceability to the original observation while providing a latent HEALPix representation suitable for DGGS-based EO data cube workflows.

Future work will focus on systematic comparisons with conventional geometric baselines, sweeps over PSF width and regularization strength, explicit frequency-band metrics for partial PSF correction, extension to additional scene types and sensors, and storage optimization for large-scale HEALPix EO archives.

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